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Designing sustainable diets that prioritize local and seasonal foods

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ABSTRACT

Food miles account for nearly 20% of global food emissions. Prioritizing the use of local foods is therefore an important opportunity to reduce the environmental impact of our food system. Designing diets that are nutritious, affordable, and adapted to local availability is challenging due to the conflicting nature of these objectives. While mathematical programming has proven effective for such trade-offs, most studies overlook seasonal ingredient availability and its cultural relevance, limiting their practical value. This study develops a mixed-integer meal planning model that integrates seasonal ingredient availability with nutritional value, environmental impact, and costs. Using Andalusia, Spain, as a case study, the framework shows that local food consumption can be increased year-round at lower cost than current diets, while meeting nutritional requirements and reducing greenhouse gas emissions. By explicitly linking seasonality, regional resource availability, and dietary shifts, this framework supports the design of culturally acceptable, resource-efficient diets and informs policies promoting sustainable food systems.

1. Introduction

Global food trade has grown rapidly in recent decades due to a reduction in transportation cost and an increasing demand for diverse foods (FAO, 2020). The distance food travels from production to consumption is measured by the concept of food miles. According to Li et al. (2022), food miles account for nearly 20% of global food emissions, which is 3.5–7.5 times higher than previous estimates, due to the justified inclusion of all upstream supply chains. Half of these emissions are associated with high-income countries, with a large share linked to the transportation of fruits and vegetables due to their high tonnages and requirement for temperature-controlled environments. Reducing food miles and increasing local and seasonal consumption is an opportunity to reduce the environmental impact of our food system.

Increasing the regional consumption of locally produced foods has been proposed as a natural solution to reduce food miles. However, systematic reviews challenge the idea that local food consumption is inherently sustainable or healthy (Stein and Santini, 2022; Enthoven and Van den Broeck, 2021). The environmental benefits of local food systems can be offset by inefficient production systems (Röös and Karlsson, 2013; Brodt et al., 2013). For instance, producing vegetables in heated greenhouses often has a higher environmental impact than growing them in open fields (Theurl et al., 2014). Nevertheless, when considering seasonal production, consuming locally grown foods can reduce greenhouse gas emissions (GHGE) (Tobarra et al., 2018; Milà i Canals et al., 2007; Burg et al., 2025). Considering the fact that food miles emissions are far higher than previously estimated, dietary change towards more plant-based foods should be coupled with an increasing use of seasonal, locally produced foods to mitigate the environmental impact of the food system (Nkonya et al., 2019; Li et al., 2022).

Designing diets that are nutritionally adequate and affordable while relying on local foods is challenging due to the conflicting nature of these objectives (Garnett, 2011; Donkor et al., 2023). Diet modeling has proven to be an effective method to formulate nutritious and affordable diets (Gazan et al., 2018; Van Dooren, 2018; Liu et al., 2024). Mathematical diet modeling applies optimization techniques to find an optimal combination of foods that best address specific objectives while adhering to constraints (Buttriss et al., 2014). However, translating the results of a diet model into practice can be difficult. A diet model usually returns a list of ingredients, but these might not work together as a culturally acceptable meal (Macdiarmid et al., 2012). In contrast, meal planning aims to improve the acceptability of proposed diets by finding combinations of meals instead of ingredients to fulfill a certain objective (Martos-Barrachina et al., 2024). The use of local foods is often implicitly encouraged in meal planning by using datasets with local recipes (Benvenuti and De Santis, 2020). However, by overlooking seasonal changes in food availability, most meal plans offer a static solution that may not be feasible all year-round. Incorporating seasonality

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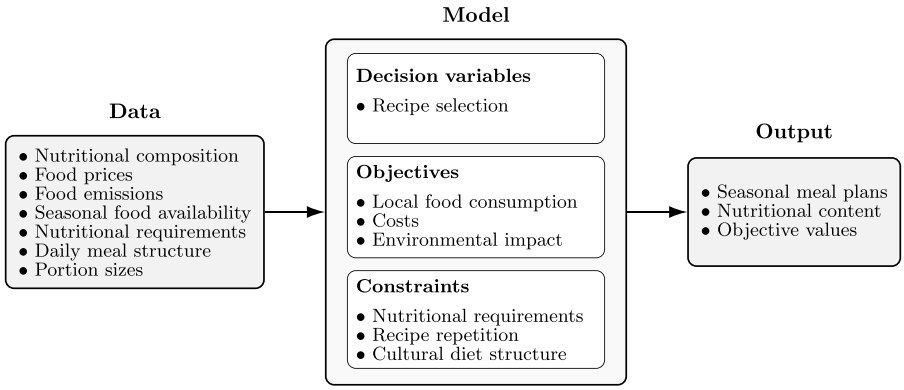


Fig. 1. Overview of the meal planning model framework main data inputs, model structure, and outputs. The model is solved for every month.

into meal planning could help stakeholders assess nutritional adequacy across the year and design dietary strategies that align with regional agricultural cycles.

Considering that food mile emissions are far higher than previously estimated, there is a need for dietary strategies that increase local food consumption without compromising nutritional adequacy, affordability, or environmental impact. This study advances existing research by introducing a meal planning framework that simultaneously considers seasonality, nutritional adequacy, cost, and GHGE. By explicitly incorporating seasonal availability of ingredients, the framework produces year-round feasible meal plans that are more practical for implementation than static approaches. Using Andalusia, Spain, as a case study, the model evaluates the extent to which locally produced foods can be integrated into nutritionally balanced diets across different seasons. The meal planning framework is adaptable to diverse food systems and can support the design of sustainable diets.

2. Research design

This study develops a meal planning framework that integrates local and seasonal food availability into diet optimization. The framework aims to find combinations of recipes that increase local consumption while ensuring nutritional adequacy, affordability, and cultural acceptability. To achieve this, a mixed-integer programming model is formulated. The model generates monthly meal plans by selecting recipes for each slot within a culturally defined meal structure, accounting for seasonal food availability. Constraints include nutritional requirements, limits on recipe repetitions, and culturally relevant customs such as the balance between specific food categories. In addition to increasing the proportion of local food in meal plans, the model can be used to minimize cost or environmental impact, or to explore trade-offs among these objectives.

A schematic overview of the model structure and required data inputs is presented in Fig. 1. The framework is demonstrated using Andalusia, Spain, as a case study. The following sections give a formal description of the model (Section 2.1) and describe the data sources and parameters used (Section 2.2).

2.1. Model

The model is formulated as a mixed-integer linear program that selects recipes for each predefined meal slot to formulate meal plans. The key decision variables $X_{r,k,d}$ indicate whether recipe r is selected for meal slot k on day d. The model is solved separately for each month m, which is held constant during each optimization. Although the model could be formulated over the full year, there are only a small number of constraints that connect the separate months. Due to seasonality in ingredient availability, the monthly problems are weakly coupled. Therefore, solving the model separately for each month provides a close approximation to a full-year formulation. Table 1 summarizes the symbols used in the model, including sets, indices, decision variables, and parameters. The following subsections present the objective functions and constraints in detail.

Table 1
Summary of notation: sets, decision variables and parameters used in the model.

Symbol	Description
Sets and indices	
$d \in D$	Days
$g \in G$	Food groups
$i \in I$	Ingredients
$I_g \subseteq I$	Ingredients that belong to group g, for each $g \in G$
$k \in K$	Meal slots (e.g., lunch starter, dinner main)
$m \in M$	Months (parameter for model runs)
$n \in N$	Nutrients
$NF \subseteq N$	Fat-soluble nutrients
$NM \subseteq N$	Macro nutrients
$NW \subseteq N$	Water-soluble nutrients
$r \in R$	Recipes
Decision variables	
$ni_{n,d}$	Nutrient intake of nutrient n on day d
$x_{r,k,d}$	1 if recipe r is selected for slot k on day d, 0 otherwise
Parameters	
$COST_{i,m}$	Price of ingredient i in month m
$FCD_{i,n}$	Nutrient amount of nutrient n per gram of ingredient i
$FGLB_g$, $FGUB_g$	Lower and upper bound for food group g in grams
$GHGE_i$	Greenhouse gas emissions of ingredient i (g CO_2e per g)
$L_{i,m}$	1 if ingredient i is locally available in month m, 0 otherwise
LB_n, UB_n	Lower and upper bound for daily intake of nutrient n
$P_{i,r}$	Quantity in grams of ingredient i in recipe r
$S_{r,k}$	1 if recipe r can be selected for slot k, 0 otherwise
T	Maximum allowed repetitions of a recipe

2.1.1. Objective functions

The primary objective of the model is to enhance the proportion of the diet that is sourced locally, while also considering cost and environmental impact. Therefore, three objectives are defined: local food consumption (f_1), monetary cost (f_2), and GHGE (f_3).

Objective f_1 represents the total weight of foods, referred to as ingredients, in the meal plan that are locally available. Here, parameter $P_{i,r}$ denotes the amount (g) of ingredient i in recipe r and $L_{i,m}$ equals 1 if ingredient i is locally available in month m, and 0 otherwise. For a given month, the total weight of the locally available foods in the meal plan is:

$$f_1(\mathbf{x};\mathbf{m}) = \sum_{r\in R}\sum_{k\in K}\sum_{d\in D}\sum_{i\in I} x_{r,k,d}\cdot P_{i,r}\cdot L_{i,m} \quad (1)$$

The cost objective, f_2, multiplies the quantity of each ingredient i by its cost in month m, $COST_{i,m}$.

$$f_2(\mathbf{x};\mathbf{m}) = \sum_{r\in R}\sum_{k\in K}\sum_{d\in D}\sum_{i\in I} x_{r,k,d}\cdot P_{i,r}\cdot COST_{i,m} \tag{2}$$

Similarly, the GHGE objective, f_3, multiplies the quantity of each ingredient i by its greenhouse gas emissions, $GHGE_i$:

$$f_3(\mathbf{x};\mathbf{m}) = \sum_{r\in R}\sum_{k\in K}\sum_{d\in D}\sum_{i\in I} x_{r,k,d}\cdot P_{i,r}\cdot GHGE_i \tag{3}$$

The objectives are normalized using min–max scaling based on reference values obtained from single-objective optimization runs, as shown in the payoff matrix in Table A.1. Local food consumption is normalized using standard min–max scaling, whereas cost and GHGE are inversely scaled such that higher normalized values correspond to lower cost and lower emissions.

For the main analyses in this study, local food consumption is optimized while including a small GHGE penalty term to avoid meal plans with excessively high emissions. The corresponding achievement function is defined as:

$$\max Z_1(\mathbf{x};\mathbf{m}) = \tilde{f_1}(\mathbf{x};\mathbf{m}) + \beta\cdot\tilde{f_3}(\mathbf{x};\mathbf{m}) \tag{4}$$

where $\tilde{f_1}$ and $\tilde{f_3}$ denote the normalized local food consumption and GHGE indicators, respectively, and $\beta = 0.1$. No extensive GHGE optimization was conducted due to data limitations.

To analyze trade-offs between local food consumption and cost, a second achievement function is defined using a weighted-sum formulation:

$$\max Z_2(\mathbf{x};\mathbf{m}) = w_1\cdot\tilde{f}_1(\mathbf{x};\mathbf{m}) + w_2\cdot\tilde{f}_2(\mathbf{x};\mathbf{m}) + \beta\cdot\tilde{f}_3(\mathbf{x};\mathbf{m}) \tag{5}$$

where different values of the weights w_1 and w_2, with $w_1 + w_2 = 1$ and $w_1, w_2 \geq 0$, are used to obtain Pareto efficient solutions.

2.1.2. Constraints

Each meal slot k on each day d has to be filled by exactly one recipe. The following constraints enforce this:

$$\sum_{r\in R} x_{r,k,d} = 1 \quad \forall k\in K, \quad \forall d\in D \tag{6}$$

Recipes r can only be selected in compatible slots k, as indicated by parameter $S_{r,k}$. For instance, yogurt cannot be selected as a dinner main. The following constraints ensure that recipes are only chosen for compatible slots:

$$x_{r,k,d} \leq S_{r,k} \quad \forall r\in R, \quad \forall k\in K, \quad \forall d\in D \tag{7}$$

Nutrient intakes are represented by variables $ni_{n,d}$, which denote the intake of nutrient n on day d. These are computed by multiplying the quantities of ingredients in the selected recipes by their nutrient content from the food composition database, $FCD_{i,n}$:

$$ni_{n,d} = \sum_{r\in R}\sum_{k\in K}\sum_{i\in I} x_{r,k,d}\cdot P_{i,r}\cdot FCD_{i,n} \quad \forall n\in N, \quad \forall d\in D \tag{8}$$

Several types of constraints are formulated to ensure adherence to nutritional guidelines. First of all, a distinction is made between water-soluble (indicated by $n\in NW$) and fat-soluble nutrients (indicated by $n\in NF$). Nutrient intakes are expressed as daily amounts ($ni_{n,d}$), but water-soluble nutrient intake recommendations have to be met on a daily level, whereas fat-soluble nutrients are assessed over the entire period. These constraints are formulated as follows, where LB_n and UB_n refer to a nutrient's lower and upper bound, respectively:

$$ni_{n,d} \geq LB_n \quad \forall d\in D, \quad \forall n\in NW \tag{9}$$

$$ni_{n,d} \leq UB_n \quad \forall d\in D, \quad \forall n\in NW \tag{10}$$

$$\sum_{d\in D} ni_{n,d} \geq LB_n\cdot|D| \quad \forall n\in NF \tag{11}$$

$$\sum_{d\in D} ni_{n,d} \leq UB_n\cdot|D| \quad \forall n\in NF \tag{12}$$

Moreover, for selected macro nutrients (indicated by $n\in NM$), intake is constrained relative to total energy intake $ni_{energy,d}$ over the planning horizon, ensuring dietary balance:

$$\sum_{d\in D} ni_{n,d} \geq \sum_{d\in D} ni_{energy,d}\cdot LB_n \quad \forall n\in NM \tag{13}$$

$$\sum_{d\in D} ni_{n,d} \leq \sum_{d\in D} ni_{energy,d}\cdot UB_n \quad \forall n\in NM \tag{14}$$

Local food-based dietary guidelines are followed for both nutritional and cultural reasons. These guidelines specify lower and upper limits on the daily consumption of ingredient groups, such as processed meats. Each day, the intake of each food group $g\in G$ has to satisfy the lower ($FGLB_g$) and upper ($FGUB_g$) bounds. The set of ingredients belonging to food group g is denoted by I_g. These constraints are formulated as:

$$\sum_{k\in K}\sum_{r\in R}\sum_{i\in I_g} x_{r,k,d}\cdot P_{i,r} \geq FGLB_g \quad \forall d\in D, \quad \forall g\in G \tag{15}$$

$$\sum_{k\in K}\sum_{r\in R}\sum_{i\in I_g} x_{r,k,d}\cdot P_{i,r} \leq FGUB_g \quad \forall d\in D, \quad \forall g\in G \tag{16}$$

To ensure a varied diet, each recipe is limited to appearing at most T times in the meal plan.

$$\sum_{d\in D}\sum_{k\in K} x_{r,k,d} \leq T \quad \forall r\in R \tag{17}$$

Additional constraints can be introduced to reflect cultural customs specific to the case study. For example, certain ingredients may be preferred over others, or variability in protein and carbohydrate sources can be enforced.

2.2. Data

This study takes Andalusia, Spain, as a case study. Accordingly, all foods, recipes, cultural dietary patterns, and nutritional guidelines used as model inputs are specific to this region.

A set of the 275 most consumed foods in Spain is obtained from the Spanish ministry of Agriculture, Fishery, and Food (abbreviated as MAPA) and these foods are classified into 14 food groups (MAPA, 2024b). Nutritional composition data were sourced from Moreiras et al. (2017), and monthly retail prices for Andalusia were collected from the food consumption panel (MAPA, 2024b). Given the absence of region-specific GHGE data for Andalusia, average national GHGE values were used to approximate emissions and to capture the relative order of magnitude of emissions across foods (Poore and Nemecek, 2018).

Local and seasonal availability of foods in Andalusia was determined using data from two main sources to capture both agricultural and fishing seasonality. Monthly marketed volumes (commercialization, which refers to products sold through local markets) for major crops were obtained from MAPA (2024a), while Alimentaria (2023) provided information on regional fishing seasons. It is assumed that a food is in its natural growing or fishing season if its monthly commercialization exceeds 10% of its total annual volume. Foods that are locally available throughout the year, such as olive oil, are considered year-round local foods, whereas foods with distinct peaks in production are treated as seasonally available.

Table 2 illustrates monthly commercialization percentages for selected foods. For example, over 74,000 tonnes of avocado are marketed in Andalusia between October and April, but the majority occurs from November to March. In October and April, less than 10% of the annual volume is commercialized, so avocado is assumed unavailable locally during those months. In contrast, cucumber is considered locally available from October to February.

Nutritional requirements are based on dietary guidelines for a moderately active woman aged 30–39 years (Moreiras et al., 2017), a commonly used reference group in nutrition research due to its moderate energy needs and relatively high micronutrient requirements. Table 3 lists absolute nutrient intake requirements, while Table 4 specifies target energy shares of macro nutrients. For the absolute nutrient

Table 2
Illustrative monthly commercialization of selected foods in Andalusia (% of annual volume). Foods are considered not in their natural growing season if monthly commercialization is below 10% (gray shading).

Food item	Commercialization (t)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Avocado	74 103	15	19	10	6						9	17	24
Cucumber	603 075	12	11	9	5	6	4			7	16	15	15
Watermelon	18 551							48	19	14	10	5	4
Pepper	714 433	11	11	13	8	7	2	2	2	8	10	13	13

Table 3
Nutritional requirements for a moderately active woman aged 30-39 years. Nutrients are classified as water-soluble or fat-soluble to indicate whether intake is evaluated daily or over the planning horizon in the model.

n	Nutrient	Lower Bound	Upper Bound	Type
1	Energy (Kcal)	1 840	2 300	n.a.[a]
2	Fiber (g)	25	–	n.a.[a]
3	Cholesterol (mg)	–	300	n.a.[a]
4	Calcium (mg)	750	–	Fat-soluble
5	Iron (mg)	14	–	Fat-soluble
6	Magnesium (mg)	330	–	n.a.[a]
7	Sodium (mg)	–	2 000	n.a.[a]
8	Potassium (mg)	2 000	–	n.a.[a]
9	Phosphorus (mg)	700	–	Fat-soluble
10	Niacin (mg)	15	–	Water-soluble
11	Folate (μg)	400	–	Water-soluble
12	Vitamin B12 (μg)	2	–	Water-soluble
13	Vitamin C (mg)	80	–	Water-soluble
14	Vitamin A (μg)	800	–	Fat-soluble
15	Vitamin D (μg)	5	–	Fat-soluble
16	Vitamin E (mg)	12	–	Fat-soluble

[a] Nutrients not classified as water- or fat-soluble; evaluated daily in the model.

Table 4
Proportion of energy required from macronutrients.

n	Macronutrient (% of Total kcal)	Lower Bound	Upper Bound
17	Protein	10	20
18	Carbohydrates	40	60
19	Sugar	–	9
20	Fat	25	40

Table 5
Average daily Food Based Dietary Guidelines (FBDG) according to Aranceta and Majem (2011).

Food Item (or Group)	Lower Bound	Upper Bound
Red Meat (g)	–	50
Processed Meat (g)	–	20
Fish (g)	40	–
Vegetables (g)	300	–
Fruit (g)	250	–
EVOO (g)	25	–
Butter (g)	–	10
Legumes (g)	6.5	–
Nuts (g)	22	–
Sweets (g)	–	9

intakes, a distinction is made between nutrients whose intake has to be met everyday (water-soluble, macro nutrients, and minerals), and nutrients whose intake has to be met over the planning horizon of the meal plan (fat-soluble). Furthermore, there are additional requirements for fat intake: unsaturated fat intakes should be at least twice that of saturated fats, and polyunsaturated fats reach at least 50% of saturated fat intake (Moreiras et al., 2017).

Food Based Dietary Guidelines (FBDG) for the Spanish and Mediterranean Diet are shown in Table 5 (Aranceta and Majem, 2011). For example, the consumption of red meat may not exceed an average of 50 grams per day in total in the meal plan. Moreover, Mediterranean Diet guidelines requires the consumption of white meat to exceed red meat, and the consumption of fish over meat. Also, meat or fish may not be repeated for lunch and dinner on the same day.

Table 6
Daily meal structure and standard portion sizes used for each meal component.

Intake	*k*	What?	Portion size (g)
Breakfast	1	Hot beverage	200
	2	Fruit or juice	150
	3	Main dish – breakfast	250
Lunch	4	Piece of bread	100
	5	Cold beverage	250
	6	Starter	250
	7	Main dish	250
	8	Dessert	150
Dinner	9	Cold beverage	250
	10	Main dish	250
	11	Dessert	150
Extra	12	Snacks	30

A database containing 230 traditional Andalusian recipes was obtained from Martos-Barrachina et al. (2022). Following Spanish traditions, it is customary to divide the daily intake into three meals (breakfast, lunch, and dinner), where lunch is the most important meal of the day. A set of intake slots k is defined, representing the usual times and components of consumption throughout the day, each with a standardized portion size (Table 6). Each day is thus divided in a sequence of slots: breakfast (three slots), lunch (five slots), dinner (three slots), and one additional slot for snacks. For instance, breakfast includes slots for a hot beverage, fruit or juice, and a main breakfast dish. In the meal plan, one recipe must be selected for each intake position k. Some recipes can be assigned for multiple contexts. For example, a banana (classified as fruit) may be selected for the breakfast fruit slot ($k = 2$) or as a dessert after dinner ($k = 8$), but never in the bread slot ($k = 4$). The database is dominated by main dishes, with approximately 130 options available for lunch and dinner.

Current food consumption data is used as a benchmark in this study. The 2023 Household Food Consumption Panel conducted by MAPA reports monthly household food purchase quantities and prices per capita for each Spanish region (MAPA, 2024b). According to the survey, on average 12.8% of total food and drink consumption takes place out

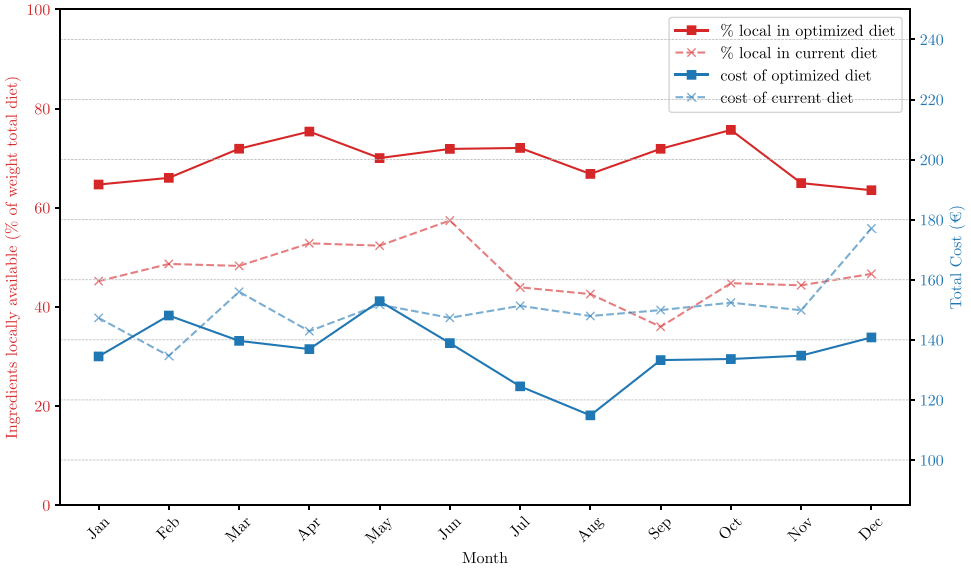


Fig. 2. Part of the diet that can be locally sourced for each month (red graph referring to the left axis) and corresponding cost for each month in euros (blue graph referring to the right axis). Optimized diet is obtained using achievement function Z_1.

of home. We therefore scale the reported in-home consumption from Andalusia to represent total consumption, and refer to this adjusted dataset as the "current diet". This dataset serves as the baseline for comparison with the optimized diets. Note that the current diet does not meet the nutritional guidelines imposed in the optimized diets. The current consumption of processed meats and sweets exceed the FBDG, whereas the consumption of fruits, vegetables, nuts, and fish do not meet the prescribed amounts.

3. Results

The model optimizes the amount of ingredients in the meal plan that can be locally sourced for each month, while also calculating associated costs and estimating GHGE. Fig. 2 shows the proportion of the diet's weight (excluding drinks) that can be locally sourced, for both the current diet and an optimized diet in which local consumption is maximized (achievement function Z_1). The corresponding values are provided in Table A.2. Local consumption refers to the use of ingredients that are either in season or available year round from local production. In the optimized diet, local consumption never falls below 64% in any month, and reaches up to 75% in April and October. On average, this represents a 67% increase in local consumption compared to the current diet (from 47% to 70%). Additionally, the figure shows the costs of both diets. The optimized diet costs an average of €136.20 in total per month (€4.54 per day), which is cheaper than the current diet at an average monthly cost of €150.60 (€5.02 per day). Some seasonal fluctuations are observed. For instance, in February, the optimized diet is slightly more expensive than the current diet due to the inclusion of high-cost seasonal ingredients such as strawberries and monkfish. In contrast, in August, inexpensive seasonal ingredients such as tomatoes results in lower costs. The optimized diet has an average carbon footprint of 2.43 kg CO_2 equivalent emissions per day (Table A.2), estimated using national life cycle assessment (LCA) data.

To illustrate the non-aggregated output of the model, Table 7 shows three days of the menu of April. As can be seen, the selected recipes follow the daily menu structure as defined in Table 6. Moreover, the model favors recipes centered around locally available ingredient in April. For instance, a stew and omelet with spinach, grilled bream, mushrooms, and juices and fruit portions of orange.

Fig. 3 shows the year-round average proportion of each food group in the current diet and diets in which local consumption is optimized for every month. In the average optimized plan, vegetables increase from 13.8% to 17.4%, fruits from 15.1% to 17.5%, legumes from 0.9% to 1.4%, and fish more than double from 1.8% to 5.3%. In contrast, meat decreases from 6.4% to 4.9% and dairy decreases from 15.2% to 5.6%. These food groups include much of the local and seasonal produce in the region and contribute to meeting nutritional requirements. The share of nuts more than triples (from 0.6% to 1.9%) and carbohydrate-rich foods such as bread rise from 8.5% to 10.3%, reflecting the fixed daily meal structure applied in the optimized meal plans, which includes mandatory portions of these items. In contrast, products from the sweets, sauces, and processed meats groups are largely absent from the optimized meal plans, as they are rarely produced locally and generally have less desirable nutrient profiles.

The average daily nutrient intakes per month are shown in Table A.3. Fig. 4 shows the number of times daily nutrient intakes are within 2% of the dietary reference values (DRV), as defined in Table 3. For nutrients with a minimum intake requirement, this corresponds to intake levels between 100% and 102% of the DRV, whereas for nutrients with a maximum intake threshold, this corresponds to intake levels between 98% and 100% of the allowed maximum. These cases therefore reflect nutrient intakes that satisfy the dietary guidelines while remaining very close to the imposed thresholds. For instance, folate intake frequently just reaches the dietary reference value during the summer months, likely due to the limited availability of folate-rich ingredients such as leafy green vegetables during this period due to the summer heat. Note that cholesterol and sodium have a maximum intake, whereas the other nutrients have a minimum intake. Even though the daily intakes are just within the guidelines on multiple days in each month of the year, the average monthly intakes of cholesterol, magnesium, and folate are well within the nutritional guidelines (Table A.3).

Fig. 5 shows Pareto fronts for April and December, illustrating the trade-off between cost and the proportion of locally sourced ingredients in the meal plan. These were obtained by combining the objective functions using weighted sums (achievement function Z_2). These months

Table 7
Example daily menus over three days. Recipes composed primarily out of locally available ingredients are marked in *italics*.

Intake	Day 1	Day 2	Day 3
Breakfast			
Hot beverage	Tea	Tea	Tea
Fruit or juice	*Orange and carrot juice*	*Orange and carrot juice*	*Orange and carrot juice*
Main dish	Bread with olive oil	Cereal bowl	Bread with olive oil
Lunch			
Piece of bread	Multigrain bread roll	Multigrain bread roll	Multigrain bread roll
Cold beverage	Water	Water	Water
Starter	*Chard scramble*	*Boiled green beans*	*Chickpea spinach stew*
Main dish	*Grilled bream and mixed salad*	Chicken skewers with fries	*Grilled mushrooms*
Dessert	*Summer fruit mix*	*Yogurt with fruit*	Kiwi
Dinner			
Cold beverage	Water	Water	Water
Main dish	*Garlic sautéed asparagus*	Meatballs in almond sauce	*Spinach omelette*
Dessert	*Orange*	*Fruit salad*	*Raspberry*
Extra			
Snacks	Pistachios	Hazelnuts	Peanuts

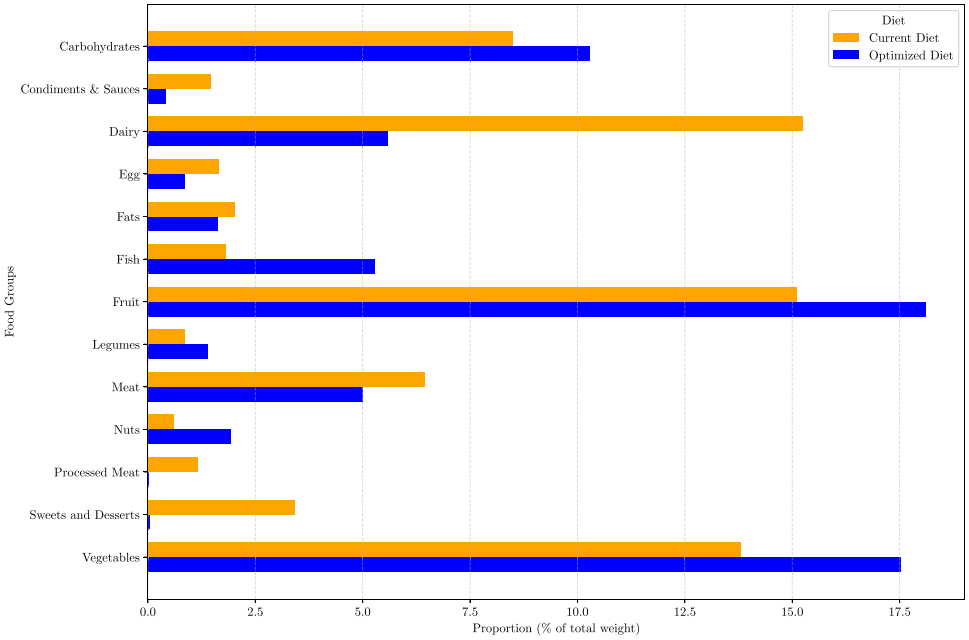


Fig. 3. Average proportion of food groups in the current diet and in diets optimized for local consumption (achievement function Z_1).

were chosen as they represent the seasonal extremes in local sourcing proportions, where April has the highest and December the lowest. The plotted points represent menus that are Pareto optimal, meaning that no other menu achieves a lower cost without reducing the local sourcing percentage, or a higher local sourcing percentage without increasing cost. Thus, for each month, the figure includes the menus that achieve the lowest cost, the highest local sourcing proportion, and the range of (optimal) compromises between them. Notably, the Pareto front for December starts at a much lower local sourcing level (52%) and extends to a higher cost (€118) compared to April. Additionally, the trade-offs in December are steeper, which means that small gains in the local sourcing proportion result in relatively large increases in cost. This highlights the greater difficulty of formulating nutritionally adequate menus with locally sourced ingredients in winter.

4. Discussion

The results of this study demonstrate that meal planning using our optimization model can formulate meal plans that increase local consumption while fulfilling nutritional requirements, at a lower cost than the current diet. Prior research shows that food miles account for nearly 20% of global food emissions, which is far higher than previous estimates (Li et al., 2022). Therefore, increasing local consumption is a promising way to decrease the environmental impact of the food system.

A key feature of the model introduced in this study is the integration of seasonal ingredient availability, resulting in meal plans that change throughout the year. In contrast to most meal planning and diet optimization studies, which typically propose static meal plans, this model accounts for fluctuations in food availability across seasons. Incorporating seasonal variation enhances the practicality of the meal plans by ensuring that ingredients are accessible when needed. It may also support cultural acceptability, since traditional diets are often aligned with local growing seasons. By explicitly combining seasonality with nutritional, cost, and environmental considerations, the model provides a framework for designing meal plans that are both feasible and regionally relevant.

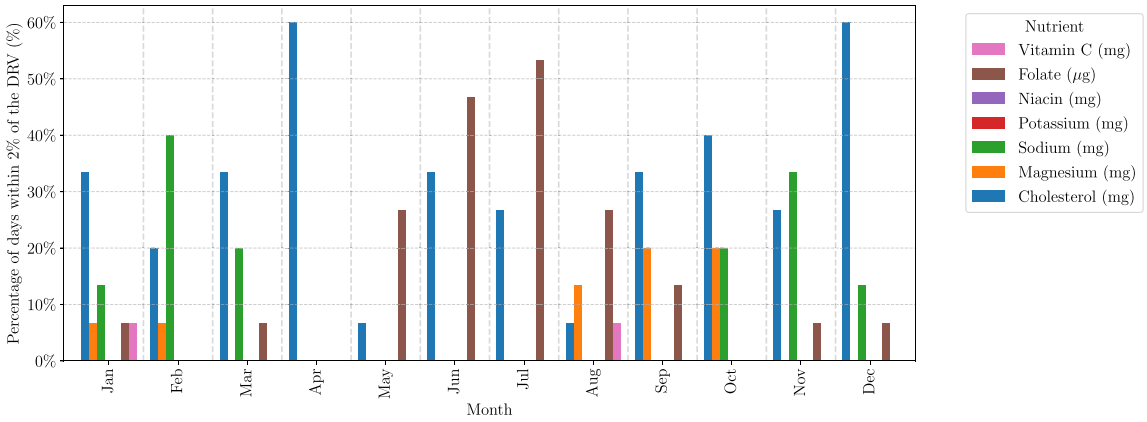


Fig. 4. Number of days per month in which nutrient intakes of the optimized diet (achievement function Z_1) are within 2% of the dietary reference values (DRV). For nutrients with minimum requirements, this corresponds to intakes between 100% and 102% of the DRV, and for nutrients with maximum thresholds, between 98% and 100% of the allowed maximum.

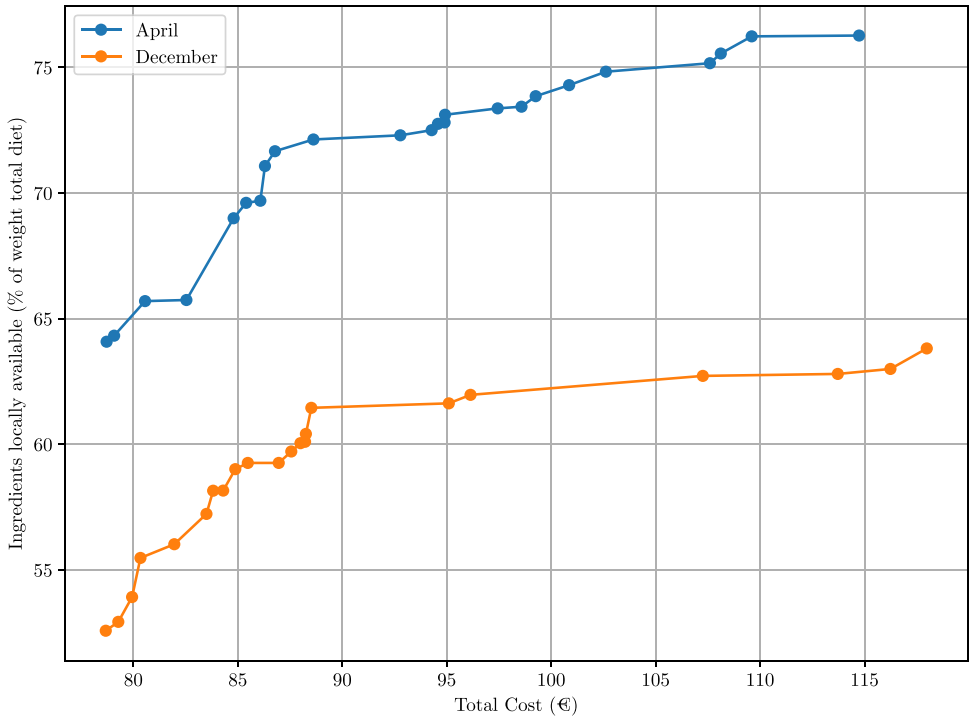


Fig. 5. Pareto fronts for April and December showing the trade-off between total cost and proportion of locally sourced ingredients in the optimized meal plans (achievement function Z_2).

Although increasing local consumption is often promoted as a sustainability strategy, it does not inherently result in lower environmental impact (Enthoven and Van den Broeck, 2021; Stein and Santini, 2022; Mundler and Rumpus, 2012). The majority of GHGE in the food supply chain are linked to the production phase (Macdiarmid, 2014). Therefore, in order to reduce GHGE, local consumption should be coupled with a dietary shift (Weber and Matthews, 2008; Nkonya et al., 2019; Burke et al., 2023). The optimized meal plans formulated in this study reflect such a shift: vegetables, fish, and fruit consumption increases, while the intake of red meat, processed meat, and dairy product declines compared to the current diet. This food pattern conforms to the dietary shifts proposed by the EAT-Lancet diet and Mediterranean diet, both of which promote health and environmental sustainability (Willett et al., 2019; Guasch-Ferré and Willett, 2021). By incorporating seasonality and local consumption, meal planning offers a promising approach to adapting these dietary guidelines to regional contexts.

National LCA data were used to approximate the GHGE associated with the optimized diets. The results suggest that the optimized meal plans (average 2.43 kg of CO_2-eq per day) generate approximately 33% lower GHGE compared with the average Spanish dietary pattern (3.62 kg of CO_2-eq per day) reported by Cambeses-Franco et al. (2022). However, the use of national averages introduces limitations, especially in capturing the regional nuances of production and distribution. Given the model's emphasis on local and seasonal ingredients, actual emissions from the optimized diets are expected to be even lower than estimated, due to reduced reliance on long-distance transport and out-of-season production. Developing regionally tailored LCA datasets

would enhance the accuracy of environmental impact assessments in future localized diet optimization studies.

The modeling framework proposed in this research is adaptable to other regions and could be used to evaluate self-sufficiency and sustainability trade-offs under different agricultural calendars and price structures. In the case of Andalusia, the results show that local consumption can be increased, while simultaneously reaching nutritional adequacy, reducing costs, and possibly lowering GHGE, in line with Spanish policy objectives for sustainable food systems (Forastero, 2023). However, outcomes depend on local production capacities. In regions with limited fresh food availability year-round, nutritional adequacy may depend on imports (Kinnunen et al., 2020). In such contexts, the model could be adapted to identify critical periods when preserved foods (e.g., frozen or canned fruits and vegetables) would be necessary to meet dietary targets (Miller and Knudson, 2014).

A critical limitation in applying the proposed meal planning framework to other regions is data availability (Kinnunen et al., 2020). The accuracy of the results depends heavily on the quality of the input data (González-García et al., 2020). Ideally, the model uses not only locally representative data but also data disaggregated by season or month. For instance, the production volumes, prices, and associated GHGE of tomatoes can vary substantially throughout the year (Zhu and Miller, 2025). In cases where region- or time-specific data are unavailable, national averages may be used as proxies, potentially in combination with sensitivity analyses to assess the robustness of the outcomes. Research on localized food systems would benefit from expanding the availability of regionally and seasonally disaggregated data (Perignon et al., 2017; de Sadeleer et al., 2024).

The recipe database and meal structure constraints used in the model can be adapted to reflect different regional and cultural dietary patterns. Whereas Andalusia has clearly defined meal slots, this may not be the case in other cultural contexts. In such cases, meal structure constraints could be simplified or partially relaxed. Likewise, the current case study applies relatively strict dietary guidelines, whereas the framework could also be adapted to improve dietary intakes under limited budgets or local food availability (Gazan et al., 2018). For example, the model could support low-income households by identifying affordable meal plans that maximize nutritional adequacy, even when not all dietary recommendations can be fully achieved. In addition, Pareto front analyses can provide a set of efficient meal plans with different trade-offs between affordability, environmental impact, nutritional adequacy, and local food consumption, allowing users to select diets that best match their priorities and preferences.

5. Conclusions

This study demonstrates that integrating seasonality into meal planning models can support nutritionally adequate and affordable diets while contributing to lower greenhouse gas emissions. By accounting for seasonal availability, the proposed framework generates meal plans that are both practical and culturally relevant, increasing the likelihood of consumer acceptance. In the case of Andalusia, the results show that local food consumption can be increased throughout the year while reaching nutritional adequacy, lowering costs, and reducing GHGE compared to the current dietary pattern. Such models can support evaluations of year-round supply feasibility and distinguish periods when preserved or imported foods are needed to meet dietary targets. When coupled with a dietary shift, increased local consumption offers a promising strategy to mitigate the environmental impact of the food system. The framework presented here contributes to this goal by simultaneously addressing nutrition, cost, emissions, and local sourcing, and provides a holistic tool for designing sustainable diets and advancing more resource-efficient food systems.

Table A.1
Payoff table used for objective normalization. For each month, three single-objective optimization runs are performed for local food maximization, cost minimization, and GHGE minimization. The resulting monthly optimal values are averaged over all months and reported in the payoff table. For cost and GHGE, inverse normalization is applied such that higher normalized values correspond to preferable outcomes.

Objective	Min	Max	Normalization
f_1: Local food consumption (g)	5048	11 939	$\tilde{f}_1(x) = \frac{f_1(x) - f_1^{\min}}{f_1^{\max} - f_1^{\min}}$
f_2: Cost (€)	39.75	75.45	$\tilde{f}_2(x) = \frac{f_2^{\max} - f_2(x)}{f_2^{\max} - f_2^{\min}}$
f_3: GHGE (g CO_2-eq)	25 345	49 275	$\tilde{f}_3(x) = \frac{f_3^{\max} - f_3(x)}{f_3^{\max} - f_3^{\min}}$

Table A.2
Monthly cost and part of the diet that can be locally sourced for the current diet and the diet in which local consumption is optimized (achievement function Z_1). For the optimized diet, CO_2-eq emissions are estimated using national life cycle assessment data.

	Current diet		Optimized diet		
Month	Local (%)	Cost (€)	Local (%)	Cost (€)	CO_2-eq (g)
Jan	45%	147.30	65%	134.48	77 939
Feb	49%	134.67	66%	148.10	74 100
Mar	48%	156.00	72%	139.66	68 046
Apr	53%	142.88	75%	136.92	73 915
May	52%	151.74	70%	152.88	74 552
Jun	57%	147.32	72%	138.92	71 394
Jul	44%	151.32	72%	124.52	80 120
Aug	43%	147.93	67%	114.86	71 205
Sep	36%	149.90	72%	133.26	81 391
Oct	45%	152.39	76%	133.62	68 120
Nov	44%	149.84	65%	134.74	60 533
Dec	47%	177.10	64%	140.80	72 519

CRediT authorship contribution statement

M.A. van Rooijen: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **F. Martos-Barrachina:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **L. Delgado-Antequera:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **J.C. Gerdessen:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: M.A. van Rooijen reports financial support was provided by European Commission. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Tables A.1–A.3.

Data availability

The source code and data used in this study are available at https://github.com/maikevr/seasonal_mpm.

Table A.3
Average daily nutrient intake of the diets in the optimized diets for local consumption (achievement function Z_1). Lower Bounds (LB) and Upper Bounds (UB) of dietary reference values are included for comparison.

n	Nutrient	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	LB	UB
1	ENER (kcal)	2075	2043	2030	2063	2123	2109	2210	2161	2180	2019	1936	2048	1840	2300
2	FIBER (g)	40	38	41	39	34	34	32	34	34	38	41	37	25	–
3	CHOL (mg)	277	244	269	275	250	257	266	237	267	265	261	293	–	300
4	CA (mg)	752	774	750	753	751	752	772	752	757	750	786	788	750	–
5	FE (mg)	30	30	31	29	17	17	17	16	28	32	31	29	14	–
6	MAGN (mg)	467	459	468	455	422	422	409	419	439	434	459	447	330	–
7	SOD (mg)	1597	1780	1649	1578	1394	1338	1211	1315	1257	1451	1673	1554	–	2000
8	POT (mg)	3950	4017	3963	3911	4104	4134	4235	3879	3884	3633	3846	3764	2000	–
9	P (mg)	1621	1532	1610	1576	1684	1612	1588	1522	1599	1377	1328	1533	700	–
10	Niacin (mg)	47	43	44	43	44	44	46	45	46	38	36	42	15	–
11	Folate (μg)	578	612	614	595	491	482	449	537	648	581	646	558	400	–
12	Vit B12 (μg)	7.74	5.35	5.03	3.21	3.54	2.86	7.13	9.31	9.07	3.80	2.53	3.63	2.00	–
13	Vit C (mg)	271	255	235	213	201	162	141	196	119	171	234	266	80	–
14	Vit A (μg)	1070	1037	1982	1879	1460	1641	806	923	802	1083	1245	1231	800	–
15	Vit D (μg)	9.34	6.25	6.37	6.07	6.84	6.64	10.99	16.61	14.27	5.05	5.09	5.06	5	–
16	Vit E (mg)	12.4	12.0	14.0	13.2	13.3	12.1	13.4	14.7	15.2	12.4	12.6	13.0	12	–
17	PROT (kcal)	415	370	379	392	409	395	398	349	380	335	329	383		
18	CHO (kcal)	874	818	826	829	852	852	886	889	884	860	799	819		
19	SUG (kcal)	283	254	275	246	265	306	331	360	327	305	304	275		
20	FAT (kcal)	707	780	748	767	799	798	866	860	849	748	706	775		
21	MUFA (g)	43	45	47	48	51	50	54	53	52	45	43	46		
22	PUFA (g)	14	16	14	15	15	15	16	17	17	14	14	16		
23	SFA (g)	15	19	16	16	17	17	19	19	18	18	15	17		

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